

UWx Real Chemistry (Nephrology) - Executive Brief

Finding a community while facing health issues plays an often overlooked, yet pivotal, part in a person's recovery. While medical consultation and medicine itself may cure a person's disease, it is the understanding and the empathy of those around them that create a holistic recovery. This community, and empathy, that is so crucial in recovery is what was found in the IgA Nephropathy patient community on Reddit. As stakeholders, it's crucial to understand the concerns and discussions that happen within this community to better treat our patients, and to better advocate for their needs. This important aspect in recovery led to our research question - What are the primary drivers of persistent negative sentiment in the IgA Nephropathy patient community, and how can this information be used in the branding strategy for our new nephrology drug treatment?

We hypothesized that if negative patient sentiment is primarily driven by struggles with clinical anxiety and steroid side effects, then our new branding must lead with empathy to address these areas and provide clarity where there is uncertainty. Since discovering that this community formed from the uncertainty that medicine leaves patients with, we also discovered that this patient uncertainty eventually leads to patient frustration— which also happens to drive community engagement.

The main frustrations that patients discussed within the Reddit community were about the physical and emotional tolls of treatment side effects, the anxiety of the disease progression, and the confusion of clinical uncertainty. All of these conversations need to be accounted for, as they are conversations that contain the clear drivers behind our patients' actions. We cannot begin to create our branding strategy without first understanding the patterns within these conversations. When looking at the data, we can see that there are clear patterns— “scared” and “lost” are some

of the top words that constantly repeat in the dataset. Other words, such as “pain”, come up with closely related words, such as Prednisone and steroids. Patients take their concerns that they reserve during medical appointments and bring them to this conversation on Reddit.

Using these key findings and insights, we can begin to understand the business and practical implications that the data directs us to. Our brand strategy needs to focus on leading with empathy for these patients that feel scared and lost, simplifying clinical uncertainty by providing clear/easy-to-understand resources, and positioning ourselves as a trusted guide. By doing so, we can create a safe environment for patients to express their concerns and feelings to us.

Methodology and Data Analysis

The primary dataset used in this analysis comes from the UW x Real Chemistry collaboration, which compiled Reddit posts related to nephrology and liver disease from July 2023 to July 2024. The dataset consists of user-generated content in which patients discuss their symptoms, treatment experiences, and healthcare decisions. While the dataset provides valuable firsthand patient perspectives, there is limited transparency regarding the exact collection methodology, such as which subreddits were used, how posts were filtered, or whether any sampling techniques were applied.

Each entry in the dataset represents an individual Reddit post and includes categorical variables such as title, sentiment, and richness. Because Reddit is an open, anonymous platform, the dataset captures organic, unsolicited patient experiences, making it particularly useful for understanding real-world concerns and behaviors outside of clinical environments. However, this also introduces variability in the quality and reliability of the information shared.

Analytical Approaches and Justification

Given the largely qualitative and categorical nature of the dataset, a mixed-methods content analysis approach was employed. First, we conducted manual coding to enhance the dataset by introducing new variables: Post Tone, Post Summary, and Post Category. This step was necessary because the existing variables did not provide sufficient depth to interpret patients' concerns meaningfully.

Posts were categorized by recurring themes such as symptoms, treatment discussions, diagnostic uncertainty, emotional expression, and decision-making processes. Tone was classified to capture the emotional context of each post (e.g., anxious, neutral, hopeful), while summaries provided concise descriptions of the content.

This approach was chosen because automated methods alone would likely miss nuanced emotional and contextual cues present in patient narratives. Manual classification allowed for a more accurate interpretation of patient experiences, which is critical when analyzing sensitive healthcare-related discussions.

Visualization Principles and Design Choices

Our poster adheres to Tufte's data-ink ratio principle by ensuring every visual element serves a functional purpose and displays necessary information. We maximized this ratio in our visualizations by using minimal graph themes and removing non-essential gridlines, allowing the patient sentiment trends to remain the focal point. To maintain data integrity, we also established a visual zero point for each chart. For instance, in our trend line chart, months with zero posts are clearly depicted to prevent data distortion, while the word frequency and bar charts maintain a

zero-baseline to accurately represent the magnitude of patient concerns. Furthermore, we leveraged the principle of similarity in our word frequency analysis by keeping formatting consistent across each category view. This facilitates pattern recognition across different sentiments and categories while reducing cognitive load.

While our hot-pink, monochromatic palette was mainly chosen for its eye-catching nature, we also applied it strategically throughout the poster and visualizations. We utilized a consistent sequential color scale across all charts to minimize cognitive load and create a cohesive visual experience for the audience. Once the viewer associates the darkest pink hue with “negative sentiment” from the first graph, they can interpret subsequent charts without needing to manually search each legend. The dark pink hue of the negative sentiment was also chosen due to its high contrast compared to the lighter hues linked to neutral and positive sentiments. This contrast immediately creates a pre-attentive attribute that naturally draws attention to the key information displayed in each chart, while the lighter hues provide necessary context and supportive aesthetics without competing for attention.

We also chose our poster layout to follow a “Z-pattern” of visual hierarchy, which aligns with how our audience naturally reads information. Arranging the graphs from left to right along the top of the poster facilitates the logical flow of our established narrative: moving from the broad frequency of sentiment to specific discussion topics, and finally to the specific voice of the patient. Placing our charts on the light pink cards was also intended to subtly separate our “data” section at the top of the poster from the “project information” section in the lower half.

To ensure our poster is accessible to a diverse audience, we prioritized text legibility and contrast as well. We utilized large, sans-serif typography for the supplemental information sections to ensure readability from a distance during presentations. Recognizing the importance

of inclusive design, we verified that our color choices for our charts remained distinct for viewers with color vision deficiencies. This was validated during our design process by testing the visualizations in grayscale, ensuring that the color contrast levels were sufficient to clearly distinguish between each sentiment.

Recommendations and Next Steps

The findings from this analysis make it clear that negative sentiment within the IgA nephropathy patient community is both persistent and deeply rooted in emotional and informational gaps. Patients are not only reacting to the clinical realities of their condition, but also to the uncertainty surrounding disease progression. Confusion about test results and the physical and emotional burden of treatment side effects are also becoming very apparent. For stakeholders, this highlights a critical disconnect between clinical innovation and patient perception. While treatments may be advancing, patients are not consistently experiencing or understanding their importance, which creates both a reputational risk and a strategic opportunity for organizations operating in the nephrology space.

To really connect with patients, companies should focus on being more understanding and caring in how they communicate. Instead of just talking about medical facts, they should try to understand how patients are feeling and tailor their communication so that they feel understood and heard. Using real stories from patients and speaking in a kind, supportive way can help build trust and make patients feel like they're being heard, not just spoken to. It is also really important to make complicated medical information easier to understand. A lot of patients get frustrated when they don't understand their test results, what is going to happen in their treatment, or what to expect from their disease. This means companies should create more educational material that is clear, easy to read, and explained in simple terms. It would be even

better if companies could come up with informational videos so patients can see what is going to happen to them. They should also guide patients through each step of their care, so they know what is happening and can make informed decisions. Companies and doctors need to know that even though there are experts on this topic, this is all new and unknown to the patients. This will not only reduce stress but also empower patients to take control of their care rather than feeling forced to comply simply because they don't understand the information. By doing this, companies can show patients that they really care about their well-being, not just their business.

Being open about the side effects of treatments is also really important. When we look at the data, we can see that people are often upset about the side effects of steroids, and they talk about how it affects them both physically and emotionally. If we can talk about these concerns, address them in an honest way, and give people practical tips on how to deal with them, it can help build trust and credibility. At the same time, we should remember that online communities where patients talk to each other are a big part of their experience. Patients often go to these online forums for advice and emotional support from people going through the same thing as them. This shows that there's an opportunity for companies to connect with them in a more genuine way. If companies work with patient advocacy groups or help start conversations in these communities, they can become a part of the support system these patients need, instead of just being an outside voice. In the end, companies should see themselves as more than just providers of treatment; they should be trusted guides for patients throughout their whole journey, offering tools, resources, and ongoing support that goes beyond just the initial treatment.

There are some challenges to think about when putting these ideas into action. For one, there are rules and regulations from the other that might limit how information is shared, so medical, legal, and business teams need to work together carefully to curate these ideas. It is also

important to find a balance between being open to potential side effects and keeping patients confident in their treatment options, which requires a thoughtful approach to messaging. Another challenge is making sure that resources are allocated properly and that the same message is being communicated across all channels, which can be especially tough for big organizations that want to focus on patient-centered care. Even with these obstacles, it is crucial to invest in a more compassionate and informative approach in order to achieve long-term success. This means being honest and transparent with patients, while also being sensitive to their needs and concerns. By doing so, healthcare providers can build trust and improve outcomes, which is essential for providing high-quality care.

Looking ahead, future research should focus on deepening the understanding of patient sentiment by segmenting audiences based on factors such as disease stage or treatment experience. Expanding data sources beyond Reddit to include surveys, interviews, and other platforms would provide a more comprehensive view of patient needs. Tracking sentiment over time will also be critical to evaluating the effectiveness of new strategies and refining approaches. Overall, the persistence of negative sentiment in this community underscores the need for a shift in how organizations engage with patients. By prioritizing empathy, clarity, and trust, stakeholders can transform patient frustration into meaningful engagement and stronger, more sustainable relationships.